

ANTHONY TROPEANO

Security Engineer, Applied Cryptography & Secure Systems

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SUMMARY

Security engineer who ships code. End-to-end builder, comfortable across the stack: zero-trust application architecture, SOC/NOC infrastructure, systems-level cryptography in Rust, and security tooling in Python and TypeScript. Founder of Defend I.T. Solutions. Summa cum laude graduate of the University of Arizona's NSA CAE-CO Cyber Operations program (4.0 GPA, December 2025). Open to security engineering, DevSecOps, or software engineering roles.

TECHNICAL SKILLS

Languages: Python, Rust, C / C++, TypeScript / JavaScript, Bash, SQL

Cryptography: Hybrid classical / post-quantum (ML-KEM, ML-DSA), Ed25519, ChaCha20-Poly1305, GPG code signing, mTLS, OIDC / OAuth2 (Zitadel)

Security Operations: Wazuh SIEM, Suricata, Zeek, Snort, Security Onion, Wireshark, tcpdump, Nmap, Nessus, OWASP ZAP, Burp Suite, Metasploit

Reverse Engineering & Forensics: Ghidra, IDA, FLARE VM, Autopsy, FTK

Infrastructure: Proxmox VE, OPNsense / pfSense, Docker, Ansible, Caddy, WireGuard, Linux hardening, network segmentation, Grafana, Prometheus

Web & Backend: React, Next.js, Hono, Bun, Node.js, Express, Drizzle ORM, REST / GraphQL, MariaDB, PostgreSQL, MongoDB

AI-Augmented Development: Agentic coding tools (Claude Code, GitHub Copilot, Codex) integrated into daily development, code review, and security-automation workflows

Frameworks & Standards: NIST CSF, NIST 800-63B, OWASP Top 10 / ASVS / API Security Top 10, MITRE ATT&CK, CISA Secure by Design

Testing & CI: TDD (Jest, PyTest, Bun), adversarial fuzzing, Git, GitHub Actions

PROFESSIONAL EXPERIENCE

Defend I.T. Solutions | Founder & Security Engineer

May 2025 – Present

Ocala, FL

- **Built an Ansible-driven SOC/NOC** across dual Proxmox nodes: Wazuh SIEM (12 agents), Suricata + Zeek NSM sensor, Grafana / Prometheus monitoring.
- **Architecting TrashScanner**, a zero-trust SaaS reference implementation targeting OWASP ASVS Level 2 and NIST 800-63B AAL2.
- **Developing DIS-Crypto**, a multi-crate Rust workspace presenting one correct-by-construction cryptographic API across four languages, with hybrid classical/post-quantum suites (ML-KEM, ML-DSA) selected so unsafe choices are unrepresentable.
- **Engineered O-Tether**, an inline security appliance for low-cost ARM hardware with stateful firewall, curated IPS, and signed-release verification.
- **Designed a two-tier OPNsense firewall topology (edge + internal core)** with network segmentation, custom DDNS, Unbound DNS, and Caddy reverse proxy with automated internal-CA certificate distribution.
- **Automated Proxmox security appliance deployment**, reducing manual provisioning from 2+ hours to 15–47 minutes.
- **Built and operate wedefendit.com** (Next.js, TypeScript): company site and security-awareness platform with in-house educational games mapping MITRE ATT&CK and the Cyber Kill Chain onto gameplay.

Remote

- Mentored undergraduates in Python programming, debugging methodology, and development-environment configuration for cybersecurity coursework.
- Identified recurring failure patterns and authored targeted reference material to improve student completion rates.

SELECTED PROJECTS

DIS-Crypto | Rust, hybrid classical/post-quantum cryptography • github.com/wedefendit/dis-crypto

- Multi-crate Rust workspace presenting one correct-by-construction cryptographic API across four languages (Rust, Python, Node, C++ via C-ABI FFI), with a suite registry that resolves hybrid classical/post-quantum suites (AES-256, ML-KEM, ML-DSA) and makes unsafe primitive or parameter choices unrepresentable.

TrashScanner | TypeScript, Bun, Hono, React, MariaDB, Zitadel OIDC, Stripe • In development

- Zero-trust SaaS reference implementation. Six-gate per-request verification (identity, freshness, device, possession, entitlement, ownership); Ed25519 JWT with single-use jti rotation; dual-path device auth (mTLS + Web Crypto challenge-response); GPG-signed code manifest verified at boot; least-privilege DB roles isolating billing writes.

O-Tether | Inline UTM appliance, ARM single-board hardware • Closed source - wedefendit.com/services/custom-solutions/o-tether

- Five-layer inline UTM appliance on low-cost ARM hardware that turns any router into a secure travel network: automatic phone-tether bridge, default-deny firewall, threat-intelligence domain/IP filtering, flow classification, and inline intrusion prevention. Runs only signed releases and is fail-closed. If protection cannot be enforced, the connection drops with it.

SIGINT | TypeScript, React 19, Bun, Canvas 2D + Web Workers, Docker, Heroku • github.com/iiTONELOC/sigint

- Real-time OSINT intelligence dashboard. Eight live data sources (aircraft, AIS vessels, seismic, fires, weather, GDELT events, news, and seasonal tropical cyclones) on an interactive globe rendered with a custom Canvas 2D + Web Worker engine.
- Cross-source correlation engine produces scored alerts and watch-mode notifications; cookie-authenticated API (HMAC-SHA256, HttpOnly); PWA with offline IndexedDB support. Live demo: sigint-5154d935429b.herokuapp.com

0xDL & pveauto | Rust • Published • crates.io/users/iiTONELOC

- 0xDL: Rust library for asynchronous file downloads with cryptographic integrity validation.
- pveauto: Rust library and CLI for automated download and verification of Proxmox VE ISO images.

threads-kernel | C • CYBV 489 coursework • github.com/iiTONELOC/threads-kernel

- Implemented core operating-system components in C within the THREADS teaching framework: thread scheduling, semaphores, mutual exclusion, message passing, interrupt handling, system calls, and basic disk I/O.
- Co-author, Escamilla et al., "Investigating Covert IPC in THREADS," Cyber Recon '25 (contributed the round-robin scheduler and core OS primitives).

EDUCATION

Nova Southeastern University | M.S., Artificial Intelligence (Data Science Concentration)

Expected 2028

University of Arizona | B.A.S., Cyber Operations (Cyber Engineering Emphasis)

December 2025

Summa Cum Laude • GPA 4.0 • NSA CAE-CO Designated Program • Outstanding Senior Award Nominee • Distinguished Undergraduate Scholar • Dean's List with Distinction

CERTIFICATIONS

NSA Cyber Operations Program Certificate, University of Arizona (December 2025)